

Alpaca Options Agent

Deterministic Risk & Statistical Validation

An architecture that refuses bad trades.

1. The Variance Problem

Over a 3.5-session evaluation window, a strategy with a true annualised Sharpe of **1.0** produces:

- A t-statistic of **0.118**
- A **54.7%** chance of positive P&L
- A **45.3%** chance of losing money

Skill accounts for just 1.4% of outcome variance on this horizon.

Any 3-day paper P&L figure is statistical noise. We designed a system that acknowledges this and survives the variance by validating first.

2. Statistical Validation Gate

We evaluated 21 candidate strategies (SPY, QQQ, IWM $\times$ 7 structures).

- **All 21 candidates were refused.**
- The closest: SPY vertical credit spread (78% win rate, Sharpe 1.67).
- Refused because the Sharpe CI lower bound remained negative (-0.32).

Moving-block bootstrap:

We corrected our bootstrap from an independent-draw model to a moving-block model, cutting the false-positive rate from **10.8%** to a calibrated **2.5%**.

3. The Proposer / Critic Architecture

The LLM never computes a number that reaches an order.

- **Proposer Agent:** Suggests regime and parameters.
- **Critic Agent:** Evaluates the proposal against our validation graveyard.

Live Veto: The Critic agent actively vetoed a proposed MSFT cash-secured put for citing no backtest evidence on a symbol that never cleared the gate.

Cost: \$0.022 per Proposer/Critic cycle.

4. Deterministic Risk Gate

Every order must pass a non-LLM Python risk gate before routing to the Alpaca MCP server.

- We implemented a batch model for structure-level capital at risk.
- **Old per-leg model:** Refused \$75,500 capital-at-risk for a SPY spread.
- **New batch model:** Correctly calculated \$423 risk and approved the trade.

Enforcement: Prevents undefined risk from reaching the broker.

5. The Demonstration Trade

Because the gate governs claims of edge, not structural safety, we executed **one** unvalidated demonstration trade.

- **Structure:** SPY defined-risk credit spread.
- **Expiry:** Friday, September 4th.
- **Exit Plan:** Closing Thursday afternoon.

Holding through Friday's nonfarm payrolls print means inflated extrinsic value and live pin risk. Managing risk is worth more than hoping for decay.

Thank You

Code public. Paper Account ID submitted.

Options quotes provided by Alpaca's free-tier Indicative Pricing Feed.